Epic Games Store Giveaways: Causal Effect on Steam Player Engagement

Nicholas Liu | 14.33 — Assignment 7

1. Revised Versions of A5 and A6

The following subsections revise each component from Assignments 5 and 6, relevant feedback from Theo is quoted in shaded boxes, followed by a brief response and the revised text.

1.1. (a) Data Sources

No specific feedback received here.

Epic Games Store giveaway history. I obtained a Kaggle dataset (Dongre, 2025) recording every free game offered through the Epic Games Store from December 2018 through 2025, comprising 541 giveaway events corresponding to 471 unique titles. The remaining 70 events are repeat appearances of titles already given away. Nine titles had the "(repeat)" suffix incorrectly applied to their *first* appearance in the source file; these were manually corrected. Repeats cluster in two patterns: *proximate* (second event within 12 months of the first, most commonly during Epic's annual holiday batches) and *distant* (>12 months).

SteamCharts monthly player data. For each EGS title matched to a Steam App ID, I scraped monthly average and peak concurrent player counts from `steamcharts.com`. SteamCharts renders historical tables server-side, so standard HTTP requests suffice. The overall panel spans July 2012 through February 2026; each game's time series begins near its Steam release date. The median game has 88 months of data.

Merge pipeline. Matching Epic game names to Steam App IDs requires reconciling inconsistent titling (subtitles, punctuation, edition suffixes). I used RapidFuzz fuzzy string matching, which auto-matched 413 of 471 unique names; the remaining 58 received manual review, of which 50 resolved to a valid Steam App ID and 8 were confirmed absent from Steam (5 Epic exclusives and 3 placeholder entries). Thirteen matched entries returned HTTP 500 errors from SteamCharts and were excluded.

Final panel. The assembled dataset contains 38,642 game–month observations across 446 games, spanning July 2012 through February 2026. Of these 446 games, 421 have SteamCharts data covering their treatment month and 336 have a full $[-12, +12]$ event window.

1.2. (b) Proposed Empirical Test

> *"Is there a reason you are not including the games that were never given away as controls? … Right now, from my understanding, your specification is a no-never-treated event study, and the lack of never treated units introduces some complexity. You might need to exclude at least two periods to avoid collinearity with your time fixed effects, and because of that, you'd need to add in the assumption that the omitted periods can be pooled. If you add never treated controls (or do a stacked event study that essentially creates never treated controls), you avoid some of that complexity. Also, are you planning to only use the observations in* $[-12, 12]$ *or will you be binning your event study at* $-12+$ *and* $12+$*?"*

Response.

Never-treated controls. I considered adding never-treated controls but ultimately did not construct an external never-treated pool for two reasons. First, there is no principled, non-selective sampling

frame for choosing which of the approximately 50,000 non-EGS Steam titles to include; any genre-, vintage-, or popularity-based criterion risks assembling a control group whose trajectories happen to be stable, biasing the comparison. Second, the not-yet-treated design sidesteps this problem because every game in my sample is eventually treated by Epic, so the control group at any point is drawn from the same population of EGS-eligible titles.

Following your suggestion, I implemented stacked DiD (Cengiz et al., 2019) and constructed cohort-specific balanced panels with *only* not-yet-treated games as controls within each cohort stack, then estimated TWFE with cohort fixed effects. This achieves the identification benefit of a never-treated group (clean controls, no negative-weighting bias) without requiring an external sample.

Collinearity and omitted periods. Because I have 80 treatment cohorts distributed across 80 distinct calendar months (December 2018 through late 2025), at any calendar month t, the panel contains games at a wide range of event-times (recently treated, approaching treatment, and far from it). This staggered structure means the calendar-month fixed effects and event-time dummies are not collinear, and omitting only the single reference period $k = -1$ should be sufficient for identification.

Binning vs. restricting. I use boundary-bin clipping: observations with $k < -12$ are binned into the $k = -12$ cell, and observations with $k > 12$ into $k = 12$. This retains all 446 games rather than the 336 with a full $[-12, +12]$ window. Interior estimates at $k \in [-11, 11]$ are unaffected; the boundary-bin estimates at $k = \pm 12$ pool over all periods beyond the window and should be read as averages. A robustness check restricted to the 336-game full-window subsample yields qualitatively identical results.

Revised specification. The primary specification is a two-way fixed-effects (TWFE) event-study regression on the full unbalanced game–month panel:

$$\ln(y_{it} + 1) = \alpha_i + \gamma_t + \sum_{\substack{k \in [-12, 12] \\ k \neq -1}} \theta_k \cdot \mathbf{1}[\tilde{k}_{it} = k] + \varepsilon_{it} \tag{1}$$

where α_i are game fixed effects, γ_t are calendar-month fixed effects, and $\tilde{k}_{it} = \text{clip}(k_{it}, -12, 12)$ is the clipped event-time indicator. The omitted category is $k = -1$. Standard errors are clustered at the game level.

As robustness checks I compare this TWFE estimate to three complementary estimators that restrict controls to not-yet-treated games: (i) LP-DiD (Dubé et al., 2023), which estimates each period-k effect using a separate local projection on not-yet-treated controls; (ii) stacked DiD (Cengiz et al., 2019), which constructs cohort-specific balanced panels with not-yet-treated controls and estimates TWFE with cohort fixed effects; and (iii) Callaway–Sant'Anna cohort ATTs (Callaway and Sant'Anna, 2021), aggregated across cohorts weighted by cohort size.

1.3. (c) Assumptions Required for Validity

No specific feedback received here.

Parallel trends. In the absence of treatment, log player counts across cohorts would have followed the same time path after conditioning on game and calendar-month fixed effects. This requires that Epic's curation team does not systematically select games at particular inflection points in their Steam lifecycle. There is, however, reason to believe this assumption is at least partially violated: the pre-period event-study coefficients for $k \in [-12, -1]$ trace a negative slope, consistent with Epic preferentially selecting titles already in lifecycle decline. This selection concern is the central

identification caveat of the design.

No anticipation. Player behavior does not shift before the giveaway month. Epic typically announces the following week's free game only one week ahead, making systematic pre-treatment anticipation on Steam unlikely.

1.4. (d) Thoughts on Testing the Assumptions

No specific feedback received here.

Pre-trend Wald test. I test $H_0 : \theta_k = 0$ for all $k \in [-12, -2]$ jointly using a Wald test. A failure to reject supports parallel trends (necessary but not sufficient). The full-sample test yields $\chi^2 = 21.77$ ($df = 11$, $p = 0.026$), a marginal rejection. For the high-traffic subsample (215 games above the pre-treatment median), the test yields $p = 0.589$, consistent with cleaner identification for more popular titles where lifecycle noise is lower.

Popularity-stratified analysis. Splitting the sample at the median pre-treatment player count diagnoses whether the pooled result is driven by noise in the lower tail. High-traffic games show larger, more precisely estimated contemporaneous spikes and cleaner pre-trends.

Randomization placebo. I permute treatment dates across all 446 games 200 times and compare the actual $|\hat{\theta}_0|$ to the permutation null distribution. The permutation p-value is $p < 0.001$: the actual $|\hat{\theta}_0| = 0.101$ exceeds the 95th percentile of the permutation null (0.038).

Estimator comparison. Agreement across TWFE, LP-DiD, stacked DiD, and Callaway–Sant'Anna on the qualitative two-phase pattern supports internal validity; the quantitative spread across estimators informs the plausible range of the causal effect.

1.5. (e) Problems Anticipated

> *"I wonder if you have any special interest in games with repeated giveaways. If they make up a tiny share of your sample and aren't particularly interesting, I would probably just exclude them."*

Response. Repeat giveaways account for only 2% of game–month observations and the repeat-event coefficient is $\hat{\delta} = 0.0001$ ($p = 0.999$), statistically and economically indistinguishable from zero. A clean-sample robustness check that entirely drops all 30 proximate-repeat games (second giveaway within 12 months of the first, <7% of games) yields main estimates nearly identical to the full sample.

Revised text.

No never-treated group. All 446 games are eventually treated, so identification rests entirely on timing variation across cohorts. If earlier cohorts (2019–2020) differ systematically from later cohorts (2022–2025) in unobserved ways, parallel trends is harder to defend. The stacked DiD and Callaway–Sant'Anna estimators partially mitigate this by restricting controls to not-yet-treated games within each cohort.

Selection of games in lifecycle decline. The pre-period event-study coefficients trace a negative slope, consistent with Epic preferentially selecting titles already on a downward Steam trajectory. Much of the observed post-giveaway decline may therefore reflect the continuation of a pre-existing trend rather than a causal consequence of the giveaway. The LP-DiD estimate should be interpreted as an upper bound on the true causal substitution effect.

Noisy low-player-count games. Of the 446 games, 101 have a pre-treatment median below 10 concurrent players. The log transformation amplifies proportional noise for near-zero values; the

popularity-stratified analysis isolates this concern.

Repeat giveaways (resolved). The 30 proximate-repeat games (second event within 12 months) are handled via a scalar repeat_event_{it} indicator. The estimate $\hat{\delta} \approx 0$ confirms no contamination; excluding these games entirely leaves main estimates unchanged.

DLC/bundle mapping. Fifteen events distributed DLC or in-game content rather than the base game. Mapping these to the base game's Steam App ID likely attenuates the estimated treatment effect for those observations.

Player counts vs. purchases. SteamCharts reports concurrent players, not purchases or unique installs. The giveaway effect on purchasing may be larger or differently timed than the player-count effect estimated here.

1.6. (f) Brief Review of Related Literature

> *"On the paper side, I'd try to explain how your analysis differs from Clarkson's."*

Response. The revised literature review now includes a dedicated paragraph explicitly distinguishing this paper from Clarkson along three dimensions: the length of the event-study window (twelve months versus contemporaneous only), the use of pre-period path to document selection on lifecycle decline, and sample breadth (446 matched EGS titles versus a curated subset).

Revised literature review.

Cross-platform spillovers and platform competition. The most directly related work is Clarkson (2025), a doctoral dissertation whose second chapter quasi-experimentally studies cross-platform spillover effects of gaming platform giveaways. Using Steam player traffic data, Clarkson finds that games given away by a competing platform (Epic) see Steam traffic increase by more than 24%. He also documents that word-of-mouth and Steam review activity increase significantly during these promotions, and that spillover effects are larger for games that leverage Steam-exclusive features (friend lists, achievements, community hubs).

The present paper differs from Clarkson's in three key respects. First, and most substantially, Clarkson's analysis focuses on the short-run contemporaneous effect; this paper estimates the full twelve-month post-treatment trajectory, documents the systematic reversal below pre-giveaway baseline that prior work has not characterized, and employs multiple heterogeneous-treatment-robust estimators (LP-DiD, stacked DiD, Callaway–Sant'Anna) to bound the plausible range of the causal effect. Second, by tracking the pre-period event-study path, this paper provides evidence that the post-giveaway decline reflects selection on lifecycle decline rather than causal substitution alone, a mechanism invisible to a contemporaneous-only design. Third, the sample matches the full EGS giveaway catalogue (446 titles matched to Steam, 2018–2025) rather than a curated subset, enabling a broader characterization of the average effect.

More broadly, the question fits into the literature on platform competition and multi-homing in two-sided markets (Rochet and Tirole, 2003). When two platforms serve the same user base (as Steam and EGS do for PC gamers), a free promotion on one platform may reallocate engagement rather than expand the aggregate market.

Tudón (2022) estimates direct network effects on Steam and finds a 0.13% own-demand increase per 1% increase in friends' demand. This social-multiplier channel provides a mechanism for the advertising effect: by expanding the addressable player community, a free copy on a rival platform may activate network complementarities that benefit the incumbent.

Nabil and Ariatmanto (2025) find that Steam significantly outperforms EGS on content variety

and ease of use, and that Epic's free game program attracts users without closing the satisfaction gap. This is consistent with a world where Steam's incumbency advantage is durable: players who obtain a game via Epic may still wish to engage with Steam's ecosystem for reviews, achievements, and community features.

Methodologically, the backbone draws on Callaway and Sant'Anna (2021) for cohort-robust staggered DiD, Cengiz et al. (2019) for stacked DiD, and Dubé et al. (2023) for LP-DiD.

1.7. (g) Summary Statistics: Table and Figure

No specific feedback received here.

Table 1 presents summary statistics for the key variables. Raw player counts are highly right-skewed: the mean monthly average of 1,584 players is more than 24 times the median of 65 players, reflecting a distribution dominated by long-tail indie titles with occasional blockbusters (e.g., GTA V, Destiny 2). The log-transformed outcome is far better behaved (mean 4.43, SD 2.36), as intended. The repeat-event indicator equals 1 in only 2% of game–month observations.

Table 1: Summary statistics: Epic giveaway panel

Variable	N	Mean	SD	Min	Median	Max
Avg. concurrent Steam players (monthly)	38,642	1,583.9	7,444.0	0.00	65.2	270,553
Peak concurrent Steam players (monthly)	38,642	3,214.1	14,861.2	0	171	527,652
$\ln(\texttt{avg_players} + 1)$ [outcome]	38,642	4.43	2.36	0.00	4.19	12.51
Repeat-giveaway window indicator (0/1)	38,642	0.02	0.15	0	0	1
Months of SteamCharts data per game	446	86.6	41.1	1	88	163

Notes: Game–month panel, 446 games, July 2012–February 2026. Player counts are in raw players as reported by SteamCharts.com. The outcome variable $\ln(\texttt{avg_players} + 1)$ uses a $+1$ offset to accommodate months with zero reported players. The repeat-giveaway window indicator equals 1 in months falling within 12 months after a game's second (or later) Epic giveaway. The bottom row is game-level ($N = 446$ unique games).

Figure 1 presents two panels characterizing the outcome variable. The left panel shows overlaid density histograms of $\ln(\texttt{avg_players} + 1)$ for pre-giveaway ($k < 0$) and post-giveaway ($k \geq 0$) months. Both distributions are roughly unimodal with a long right tail; the large mass near zero reflects the many low-traffic titles. The right panel plots the unconditional mean of the outcome at each event-time $k \in [-12, 12]$: the pre-period mean is approximately flat, providing preliminary (unconditional) support for the parallel trends assumption, and a sharp uptick is visible at $k = 0$ with partial persistence into $k = 1$.

Figure 2 provides three additional descriptive panels. Panel (a) shows the roughly bimodal pre-treatment distribution of log players across games, with a large cluster of low-traffic titles and a smaller cluster of popular ones. Panel (b) shows that treatment cohorts are broadly evenly distributed across 2019–2023, providing the timing variation needed for staggered DiD. Panel (c) shows that approximately 10% of games in most cohort years have a proximate repeat giveaway.

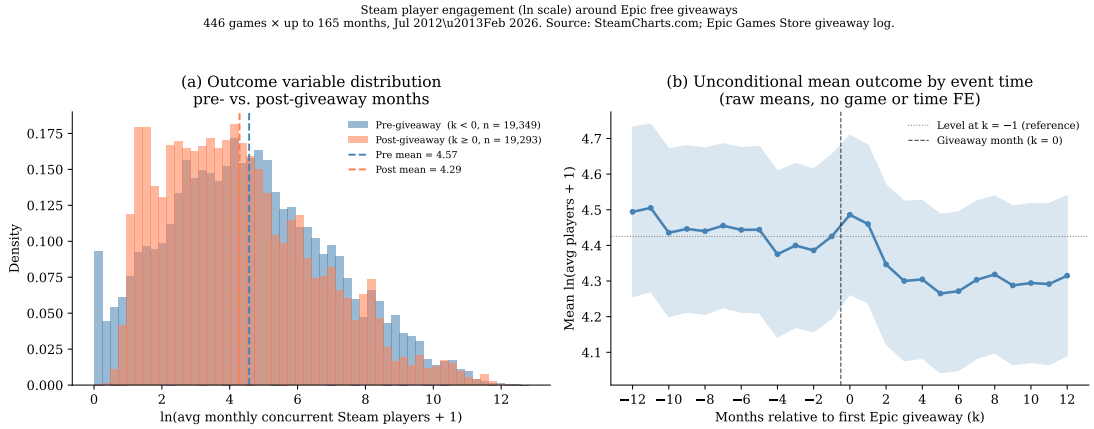

Figure 1: Steam player engagement (log scale) around Epic free giveaways. **Left:** Density histogram of $\ln(\texttt{avg_players} + 1)$ for pre-giveaway months ($k < 0$, blue) and post-giveaway months ($k \geq 0$, orange), pooled across all 446 games. **Right:** Unconditional mean (with 95% CI) of the outcome by event-time $k \in [-12, 12]$; the dashed vertical line marks the giveaway month ($k = 0$). Both panels use 38,642 game–month observations. *Source:* SteamCharts.com; Epic Games Store giveaway log (Dongre, 2025).

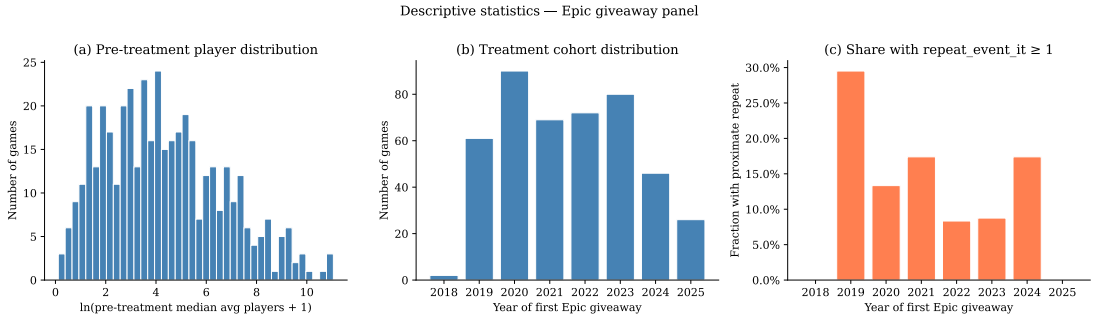

Figure 2: Panel structure and descriptive statistics. **(a)** Histogram of $\ln(\text{pre-treatment median avg players} + 1)$ across 446 games. **(b)** Number of games with first Epic giveaway in each calendar year (2018–2025). **(c)** Fraction of games with a proximate repeat giveaway (second event within 12 months of first) by cohort year. *Source:* SteamCharts.com; Epic Games Store giveaway log (Dongre, 2025).

2. Paper Outline

1. **Introduction**: Motivating anecdote (Blood West case, Reddit thread); research question (does an EGS giveaway increase or decrease Steam engagement?); preview of two-phase finding (short-run spike, medium-run reversal); brief description of identification strategy and data; contributions.

2. **Literature Review**

   - Cross-platform spillovers: Clarkson (2025) contemporaneous effect; differences from present paper
   - Network effects and social multipliers: Tudon (2022)
   - Platform satisfaction and incumbency: Nabil & Ariatmanto (2025)
   - Staggered DiD methodology: Callaway & Sant'Anna (2021), Dubé et al. (2023) LP-DiD, Cengiz et al. (2019) stacked DiD

3. **Data**

- Epic Games Store giveaway history (Kaggle, 541 events, 471 titles)
- SteamCharts monthly player data (scraping, panel structure)
- Merge pipeline (RapidFuzz matching, 446-game final panel)
- Summary statistics; data limitations

4. **Empirical Method**
   - Identification strategy (not-yet-treated design, staggered timing)
   - Pooled TWFE specification; identification in no-never-treated design; boundary-bin clipping
   - Separated specification: first vs. repeat giveaways
   - Robustness estimators: LP-DiD, stacked DiD, Callaway–Sant'Anna
   - Additional checks: popularity stratification, randomization placebo
5. **Results**
   - Main TWFE event study: spike at $k = 0$, reversal from $k = 2$
   - Pre-trend Wald test (full sample $p = 0.026$; high-traffic $p = 0.589$)
   - Estimator comparison: TWFE, LP-DiD, stacked DiD (quantitative spread; qualitative agreement)
   - Separated specification: $\hat{\delta} \approx 0$ for repeats
   - Popularity-stratified analysis
   - Callaway–Sant'Anna cohort ATTs
   - Randomization placebo ($p < 0.001$)
6. **Discussion**
   - Contemporaneous spike: advertising mechanism; comparison to Clarkson
   - Medium-run reversal: selection on lifecycle decline (primary explanation); direct substitution; counterfactual demand exhaustion
   - Null repeat-giveaway effect
   - Platform strategy implications for developers and Epic
   - Limitations: no never-treated group, selection concern, player counts vs. purchases, DLC mapping, genre heterogeneity
7. **Conclusion**: Summary of two-phase finding; primary mechanism (lifecycle selection > causal substitution); implications for platform strategy; directions for future work (single- vs. multiplayer split; Steam sales data; Epic-side engagement data).

3. Results

3.1. Main TWFE Event Study

Table 2 reports selected TWFE event-study coefficients ($\hat{\theta}_k$, Equation (1)) with standard errors clustered by game. The pre-period coefficients for $k \in [-5, -2]$ are small and individually insignificant, consistent with approximate parallel trends over the near-term pre-window. In the giveaway

month ($k = 0$), the coefficient is $\hat{\theta}_0 = 0.101$ (SE $= 0.022$, $p < 0.001$), implying approximately a 10.6% increase in monthly average concurrent Steam players.

The spike is short-lived. By $k = 2$ the coefficient turns negative (-0.063, $p = 0.005$), and the decline accelerates through the full window. By $k = 12$ the estimated coefficient is -0.267 (SE $= 0.053$, $p < 0.001$), implying approximately 23% fewer players relative to the pre-giveaway baseline. The R^2 of the game–month regression is 0.884 (within $R^2 = 0.008$), confirming that most variation in log players is accounted for by game-level and time fixed effects, with the event-study coefficients capturing the residual treatment effect.

Table 2: Selected TWFE event-study coefficients (pooled specification, Eq. (1))

Event time k	Estimate	SE	p-value
-5	0.005	0.025	0.833
-4	-0.003	0.024	0.915
-3	-0.009	0.025	0.710
-2	-0.011	0.022	0.608
0	$\mathbf{0.101}^{***}$	0.022	<0.001
1	0.045^{*}	0.022	0.047
2	-0.063^{**}	0.023	0.005
3	-0.097^{***}	0.024	<0.001
6	-0.160^{***}	0.027	<0.001
9	-0.178^{***}	0.031	<0.001
12	-0.267^{***}	0.053	<0.001

Notes: Dependent variable: $\ln(\texttt{avg_players} + 1)$. Game and calendar-month fixed effects; standard errors clustered by game ($N = 38{,}641$; 446 games). Omitted period: $k = -1$. $R^2 = 0.884$; within $R^2 = 0.008$. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

3.2. Estimator Comparison

Figure 3 overlays the TWFE, LP-DiD (Dubé et al., 2023), and stacked DiD (Cengiz et al., 2019) event-study estimates. All three estimators agree on the qualitative two-phase pattern (a positive spike at $k = 0$ followed by a sustained and statistically significant reversal), but they differ quantitatively in ways that are informative about the range of the true causal effect.

At $k = 0$, TWFE yields 0.101 (95% CI [0.058, 0.144]), LP-DiD yields 0.131 (CI [0.088, 0.173]), and stacked DiD yields 0.047 (CI [0.032, 0.063]). The stacked DiD estimate, which constructs cohort-balanced panels and uses only not-yet-treated controls, is likely the most conservative and is the preferred estimate of the contemporaneous effect ($\approx$5% increase). TWFE and LP-DiD exceed stacked DiD, consistent with cohort-composition bias in the former.

In the medium run, all three estimators show statistically significant declines from $k = 3$ or $k = 5$ onward. At $k = 10$, the estimates span from LP-DiD (-0.121, CI [-0.191, -0.051]) to stacked DiD (-0.164, CI [-0.201, -0.127]) to TWFE (-0.199, CI [-0.259, -0.138]). The LP-DiD estimate, which conditions most directly on pre-treatment trends by using only not-yet-treated controls at each period, provides the most conservative medium-run estimate and should be interpreted as an upper bound on the true causal substitution effect, because the pre-period negative slope indicates that treated games were already declining before the giveaway.

The pre-period path ($k \in [-12, -1]$) traces a downward slope across all three estimators, providing the most direct evidence of Epic's selection of titles already in lifecycle decline. This pre-existing trajectory likely accounts for a meaningful share of the post-giveaway reversal: the approximately

0.08 log-point gap between TWFE and LP-DiD at $k = 10$ reflects the portion of the observed reversal that LP-DiD attributes to this pre-existing selection rather than to the giveaway itself.

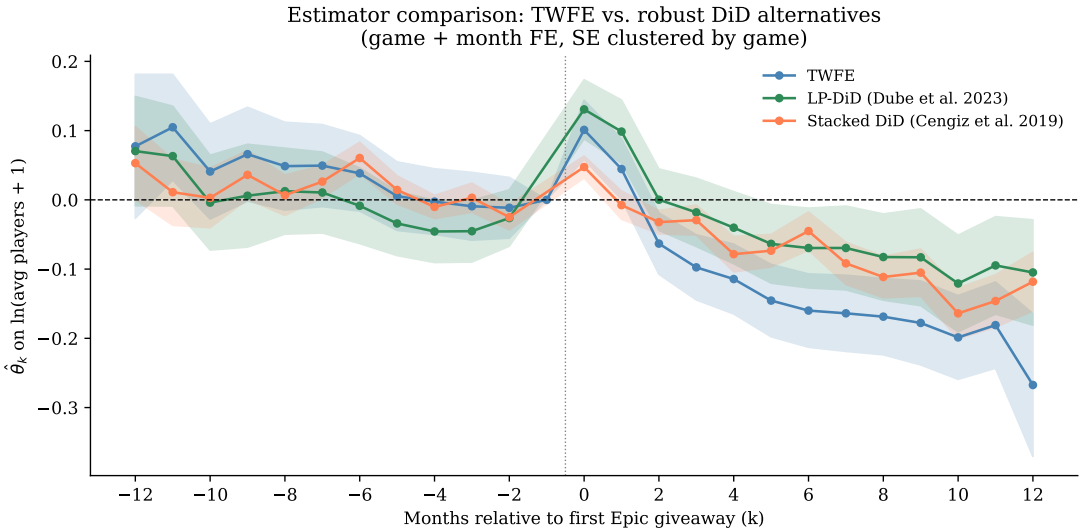

Figure 3: Event-study estimates: TWFE, LP-DiD (Dubé et al., 2023), and stacked DiD (Cengiz et al., 2019). Outcome: $\ln(\texttt{avg_players} + 1)$. LP-DiD uses only not-yet-treated controls at each period k independently. Stacked DiD constructs cohort-specific balanced panels with not-yet-treated controls, weighted by $1/$number of stacks per game. Shaded bands are 95% confidence intervals (standard errors clustered by game). Reference period $k = -1$. $N = 38{,}641$ game–month observations; 446 games.

3.3. Economic Significance and Developer Calculus

Developer calculus

The developer calculus is cleaner than it first appears. The pre-existing lifecycle decline is irrelevant to the decision: it would have happened regardless of whether the developer signs the deal. The substitution component is also not a net developer cost: a player who migrates from Steam to Epic is still playing the game, and Epic's 12% revenue cut is more favorable than Steam's 30%. The deal therefore reduces to two clearly positive terms: the licensing fee and the advertising spike.

The advertising benefit is real. We estimate a +5–14% Steam engagement boost at $k = 0$ depending on estimator. Clarkson (2025) shows Steam reviews spike alongside concurrent players during giveaways, feeding Steam's recommendation algorithm and generating downstream sales beyond our 12-month window. The Blood West case (FRVR Blog, 2026), a 200% Steam sales spike from a single EGS giveaway, suggests the purchase-conversion effect can be far larger than the engagement effect we estimate. For developers, accepting the EGS deal is almost always the correct decision.

Should Steam be concerned about EGS giveaways?

No. The giveaway program is a net positive for Steam. The $k = 0$ spike accrues directly to Steam in the form of increased concurrent players, store page traffic, and reviews. Steam's position as the primary discovery and community layer for PC gaming means that the media coverage generated by an EGS giveaway announcement funnels curious players toward Steam, not away from it. The medium-run concurrent-player decline is mostly pre-existing lifecycle decline in titles that were already losing Steam players before the giveaway; Epic is not causing Steam games to lose players,

it is selecting games that were already losing them. The residual causal substitution component reflects marginal players on declining titles, not the engaged users who sustain Steam's social ecosystem. The evidence that the advertising effect occurs on Steam and that even free distribution through a rival platform drives players to Steam's community infrastructure is consistent with Steam's competitive moat being durable.

Does the giveaway program benefit Epic?

The more interesting and harder question is whether the giveaway program benefits Epic. Epic bears the full licensing cost but receives none of the Steam advertising benefit our results document: the $k = 0$ spike is in Steam concurrent players, Steam reviews increase, and the discovery activity generated by the announcement happens on Steam's platform. What Epic plausibly gains instead is EGS user acquisition: players who install the launcher and become habitual EGS shoppers. Whether this actually happens is not answerable with our data; we observe Steam engagement but have no window into EGS-side behavior. Our results do, however, raise a pointed question: if the primary measurable effect of Epic's giveaway spending is to advertise games on Steam, Epic is paying licensing fees to benefit a competitor's platform. Whether durable EGS user acquisition offsets this is the most important open question for evaluating the program's long-run strategic rationale.

References

Callaway, B. and Sant'Anna, P.H.C. (2021). "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225(2), 200–230.

Cengiz, D., Dubé, A., Lindner, A., and Zipperer, B. (2019). "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics*, 134(3), 1405–1454.

Clarkson, T.C. (2025). "Essays on Platform Operations." Doctoral dissertation, Moore School of Business, University of South Carolina.

Dongre, P. (2025). "Epic Games Free Giveaway History 2018–2025." Kaggle dataset.

Dubé, A., Girardi, D., Jordà, O., and Taylor, A.M. (2023). "A Local Projections Approach to Difference-in-Differences Event Studies." NBER Working Paper No. 31184.

Nabil, M. and Ariatmanto, D. (2025). "Analysis of User Satisfaction and Preferences in Choosing a Digital Game Platform." *G-Tech: Jurnal Teknologi Terapan*, 9(3), 1166–1175.

Rochet, J.-C. and Tirole, J. (2003). "Platform Competition in Two-Sided Markets." *Journal of the European Economic Association*, 1(4), 990–1029.

Tudón, J. (2022). "Distilling Network Effects from Steam." *Quantitative Marketing and Economics*, 20, 293–312.

FRVR Blog (2026). "Epic Game Store's Free Giveaways Just Cause a Huge Spike in Steam Sales, Reveals New Blood CEO." *FRVR Blog*, January 20, 2026.

Note on AI usage: I used AI to assist with drafting and formatting this document in LaTeX.